

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to
Modernize the Electric Grid for a High
Distributed Energy Resources Future.

Rulemaking 21-06-017
(Filed June 24, 2021)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE
ASSIGNED COMMISSIONER'S RULING ISSUING QUESTIONS ON THE
ELECTRIFICATION IMPACT STUDY PART 2 FINAL REPORTS**

Zach Woogen
Executive Director
Vehicle-Grid Integration Council
1401 21st St., Suite 5409
Sacramento, California 95811
Telephone: (510) 665-7811
Email: vgicregulatory@vgicouncil.org

Grace Pratt
Caliber Strategies
801 K Street, Suite 2800
Sacramento, CA 95814
Telephone: (818) 264-6437
Email: grace@caliberstrat.com

Date: May 29, 2026

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments on the *Assigned Commissioner’s Ruling Issuing Questions on the Electrification Impact Study Part 2 Final Reports* issued by Commissioner Darcie Houck on May 8, 2026.

I. INTRODUCTION.

VGIC appreciates the opportunity to comment on the Assigned Commissioner’s Ruling Issuing Questions on the Electrification Impact Study Part 2 (“EIS2”) Final Reports. The EIS2 studies provide important evidence demonstrating that electrification, when paired with demand flexibility strategies, can reduce distribution infrastructure costs and provide benefits to ratepayers. For example, Pacific Gas and Electric (“PG&E”) found that “Electrification growth may provide downward pressure on distribution rates by as much as 25% by 2040.”¹ The Public Advocates Office at the California Public Utilities Commission (“Cal Advocates”) similarly found in its Distribution Grid Electrification Model 2025 report that electrification can reduce rates by 0.2 to

¹ PG&E EIS2 at 14.

4.5 cents per kilowatt-hour by 2040.² With California facing continued affordability pressures while pursuing ambitious climate goals, these findings underscore the importance of investing in electrification strategies that minimize system costs and maximize grid efficiency.

The EIS2 studies also highlight the significant value of electric vehicle (“EV”) demand flexibility and vehicle-grid integration (“VGI”) in reducing future distribution infrastructure investment needs. Across the studies, EV-enabled load flexibility emerges as one of the largest sources of potential load reduction and avoided infrastructure costs. In particular, PG&E’s analysis demonstrates that active managed charging and bidirectional EV capabilities can provide substantial peak load reductions that help defer or reduce traditional grid upgrades. VGIC therefore provides the following comments:

- The Commission should begin incorporating EIS2 findings into future Distribution Planning and Execution Process (“DPEP”) cycles and related distribution planning activities.
- Additional analysis is warranted regarding the contribution of specific demand flexibility resources, particularly managed EV charging and bidirectional charging systems.
- Southern California Edison (“SCE”) and San Diego Gas and Electric (“SDG&E”) should include bidirectional EV capabilities in future demand flexibility and distribution planning evaluations consistent with PG&E’s approach.

II. RESPONSES TO RULING QUESTIONS

1. How should the findings of the EIS2 inform the yearly DPEP?

² Cal Advocates, Distribution Grid Electrification Model 2025 at p.15. Available at: <https://www.publicadvocates.cpuc.ca.gov/press-room/reports-and-analyses/distribution-grid-electrification-model-2025>

The EIS2 reports provide some of the clearest evidence to date that electrification, when paired with coordinated demand flexibility strategies, can reduce distribution system costs and lower ratepayer impacts. The studies therefore should not remain purely exploratory exercises. Instead, the Commission should begin incorporating EIS2 findings into yearly DPEP planning assumptions, particularly in areas related to transportation electrification and flexible load management.

Rather than planning solely around unmanaged peak demand growth, the DPEP should increasingly evaluate opportunities to use managed charging, coordinated distributed energy resource (“DER”) operations, and bidirectional EV capabilities to defer or reduce traditional infrastructure upgrades where cost-effective. This is particularly important because transportation electrification is expected to become one of the largest drivers of future distribution load growth in California.

4. The EIS2 models enhanced demand flexibility and the coordinated management of DERs as a cost-saving tool to reduce future distribution infrastructure investment needs. Are the underlying assumptions of demand flexibility (including the scale of DER participation, load flexibility response rates, and timing of deployment) sufficiently supported by current data to be reliable as inputs for distribution planning and cost forecasting?

a. If yes, how will the utilities incorporate these assumptions into future DPEP cycles?

b. If not, what must utilities demonstrate to establish that those projections are sufficiently reliable, and what time frame expectations should the Commission establish?

Generally, the Final EIS2 studies demonstrate that demand flexibility should begin informing future distribution planning efforts, particularly because the studies consistently show that coordinated load management can reduce future infrastructure investment needs. However,

additional granularity regarding the contribution of specific demand flexibility resources would help the Commission and utilities determine how best to incorporate these findings into future planning processes.

All three IOUs modeled multiple forms of demand flexibility, including building electrification, behind-the-meter solar and stationary storage, and vehicle-grid integration (“VGI”). However, only PG&E identifies in detail how each of these demand flexibility resources contributes to overall load reduction in the enhanced demand flexibility scenario.³ By contrast, SCE and SDG&E do not provide comparable detail regarding how individual technologies or strategies contribute to the approximately \$1.38 billion and \$697 million in modeled savings associated with their respective demand flexibility scenarios.⁴

Providing more granular results is important for two reasons. First, it helps evaluate whether specific assumptions are sufficiently validated for distribution planning purposes. Second, it provides important insight into which demand flexibility strategies are likely to provide the greatest ratepayer value and infrastructure cost savings.

In particular, active managed EV charging has the potential to defer significant amounts of distribution infrastructure investment. PG&E found that active EV charging management could provide 1,007 MW of peak load reduction by 2040.⁵ While SCE and SDG&E modeled some elements of active managed charging, neither utility clearly quantified the specific contribution of managed charging to overall load reductions or avoided costs.⁶ Additional analysis on the impacts of managed charging would be valuable given that Cal Advocates found that shifting EV charging

³ See PG&E EIS2 at Table 6 showing MW contributions of different technologies to coincident load flexibility.

⁴ SCE EIS2 at 5; SDG&E EIS2 at 1.

⁵ PG&E EIS2 at Table 6.

⁶ SCE EIS2 at 37-38; SDG&E EIS2 at 13-16.

to manage distribution circuit load could provide between \$5 billion and \$18 billion in cost savings by 2040.⁷

6. Does EIS2 reveal any methodological or analytical limitations that should be considered before applying its findings in regulatory planning processes?

a. What factors need to be considered or further validated before incorporating EIS2 findings directly into future planning processes?

b. Please identify the specific aspects of the study where additional clarification or analysis may be warranted.

Additional analysis from SCE and SDG&E is warranted regarding the potential for bidirectional charging systems to provide demand flexibility and reduce future distribution infrastructure needs. Neither SCE nor SDG&E included bidirectional charging systems as a source of demand flexibility capable of serving onsite load or exporting energy back to the grid.

By contrast, **PG&E found that bidirectional charging systems could provide 859 MW of the utility’s estimated 2,770 MW of load reduction by 2040, representing more than 30% of PG&E’s total estimated load flexibility potential.**⁸ Given the scale of this potential resource, future demand flexibility and distribution planning evaluations should include bidirectional charging capabilities as part of the analysis.

This is particularly important because bidirectional EV and charging equipment availability is rapidly increasing across both light-duty and medium- and heavy-duty transportation sectors. Future studies should therefore continue refining assumptions of bidirectional charging adoption and evaluating how these resources can support localized distribution planning needs.

⁷ Cal Advocates, Distribution Grid Electrification Model 2025 at p.83. Available at: <https://www.publicadvocates.cpuc.ca.gov/press-room/reports-and-analyses/distribution-grid-electrification-model-2025>

⁸ PG&E EIS2 at Table 6.

III. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the EIS2 studies. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

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